

Erdős Problem #667

EP-667

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $p, q \geq 1$ be fixed integers. We define $H(n) = H(N; p, q)$ to be the largest m such that any graph on n vertices where every set of p vertices spans at least q edges must contain a complete graph on m vertices.

Is $c(p, q) = \liminf \frac{\log H(n)}{\log n}$ a strictly increasing function of q for $1 \leq q \leq \binom{p-1}{2} + 1$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>